

LionX Migration

Replatforming MNY to a cloud-native foundation on AWS.



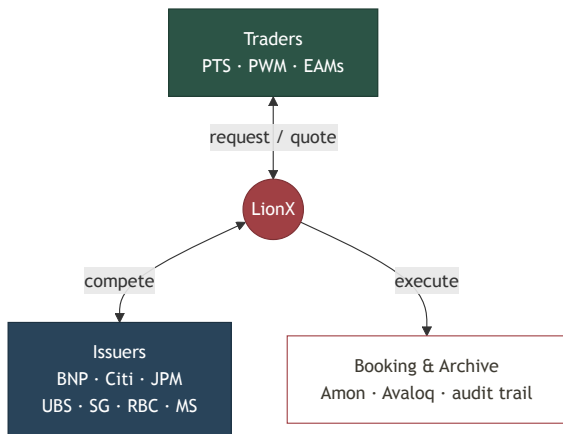
On-prem today



AWS cloud-native

What is LionX

Pictet's trading platform for structured products (internal name: MNY).



Connect

one protocol to every issuer



Quote

issuers compete, best execution



Execute

routed to internal booking (Amon, Avaloq)



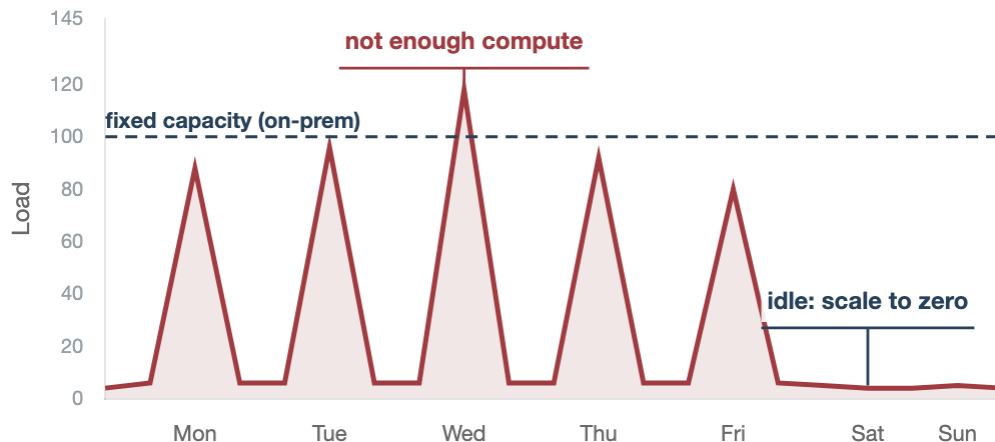
Archive

every quote and trade, full audit trail

One interface for traders. One integration point for issuers, across PTS, PWM and EAMs.

Why move to AWS

Our load peaks on market data and sits near zero off-hours and weekends.



Cloud-native

AWS flexibility, autoscaling on peaks



Full control

over development, operations and infrastructure



Scalable & auditable

a foundation built for growth and compliance



Continuity first

replatform as-is, then improve over time

Fixed on-prem capacity sits idle most of the week, yet still falls short at peaks. AWS scales up on spikes and down to zero when idle.

What holds us back today

The on-prem stack works, but it shows its limits.



Single-node MongoDB

reliability and architecture at risk



No horizontal scaling

several components cannot absorb peaks



Stop-the-world deploys

PROD is stopped to upgrade, no blue/green

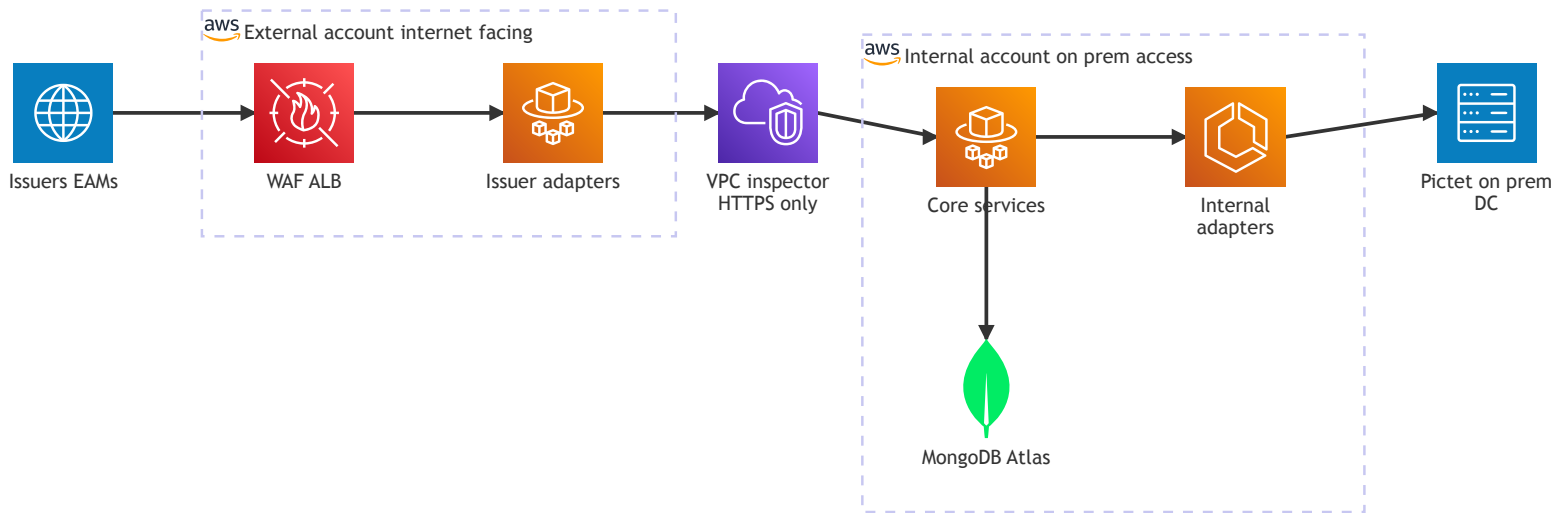


Complex pipeline

pull-based hybrid, Puppet-driven

Target architecture, step 1: replatform

Not a lift-and-shift: we retire the hybrid setup for managed services (Atlas, ALB) and a single deployment model.



Issuers and EAMs enter through the external account. PTS and PWM traders reach the core from on-prem, over the private link.

Shared across both accounts



CloudWatch



Secrets Manager



ACM PCA

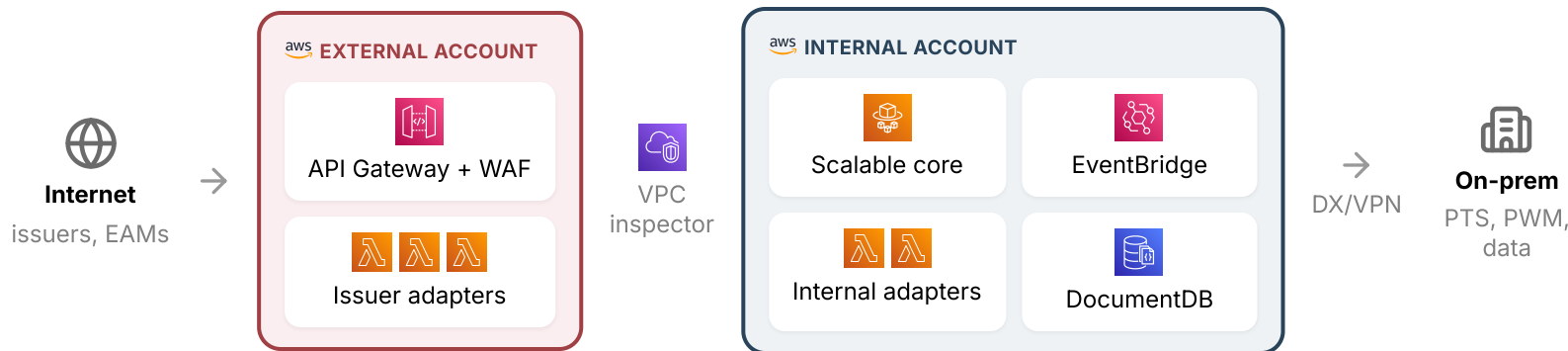


S3 snapshots



Target architecture, step 2: cloud-native

After the refactor: Lambda adapters behind an API gateway, a simplified scalable core, no in-memory grid.



Issuers and EAMs hit the API gateway; PTS and PWM traders reach the core from on-prem. Adapters run as Lambdas, the core scales horizontally, and the in-memory grid is gone.

Shared across both accounts

EventBridge

CloudWatch

Secrets Manager

S3

A two-phase migration

Business continuity first, cloud-native gains next.

PHASE 1: REPLATFORM



Iso-functional, simpler stack

same app behavior; managed services replace the hybrid setup



Parity

performance, security, resilience and audit at least equivalent



Modern operations

single push-based CI/CD, IaC, blue/green deploys

PHASE 2: LEVERAGE THE CLOUD



Elasticity

autoscaling on demand; scale down to 20% on nights and weekends



High availability

multi-AZ, self-healing, automated DR and failover tests



Serverless and cloud-native

FaaS adapters on Lambda, DocumentDB, no in-memory grid

Roadmap

Migrate first to reach parity, then modernize. Phased from 2026 to 2028.

MIGRATE

Preparation and POC

Replatforming on INTG

Cutover INTG

Roll out all envs to PROD

Migration complete

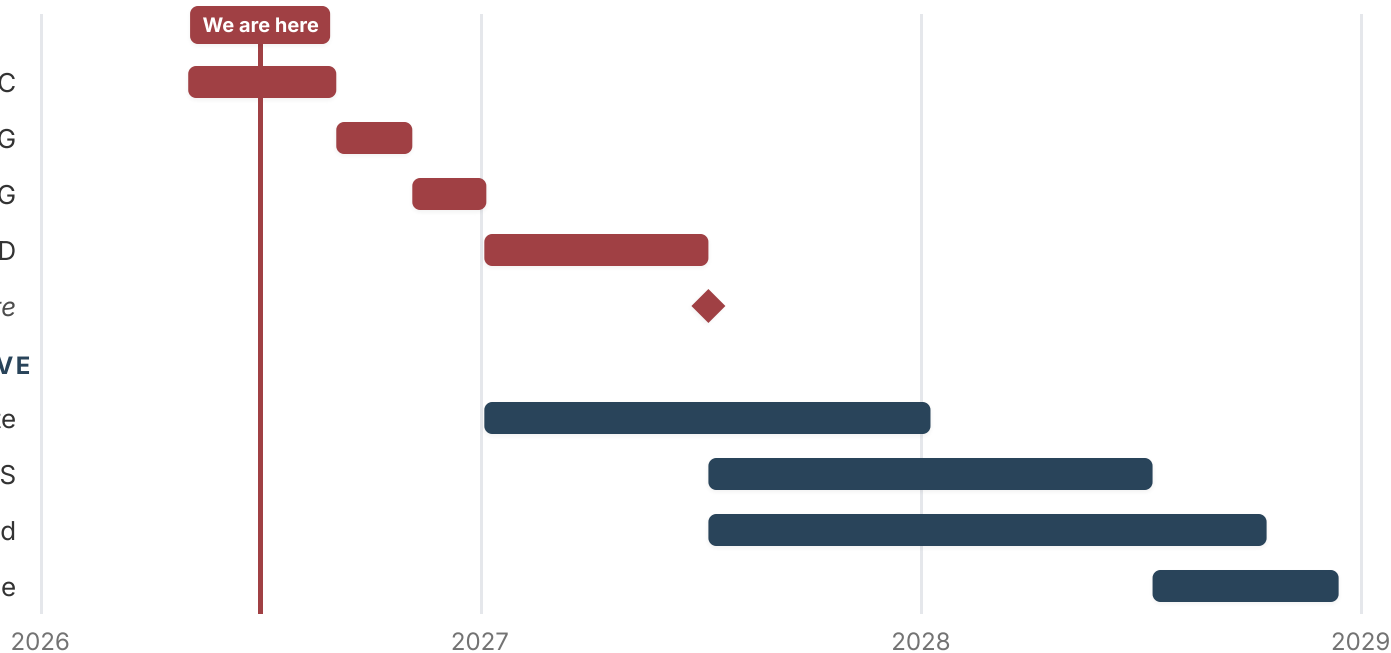
MAKE LIONX CLOUD NATIVE

Horizontally scalable compute

Function workloads to FaaS

Replace in-memory data grid

Run and optimize



Production on AWS by mid 2027, fully cloud-native by end 2028.

Risks we are managing

Known upfront, with mitigation.



Shared Pictet services

their maturity paces us; possible temporary dual monitoring



Network complexity

on-prem links, issuer connectivity, firewall inspection



Replatforming surprises

different IO, network and timeout behavior in the cloud



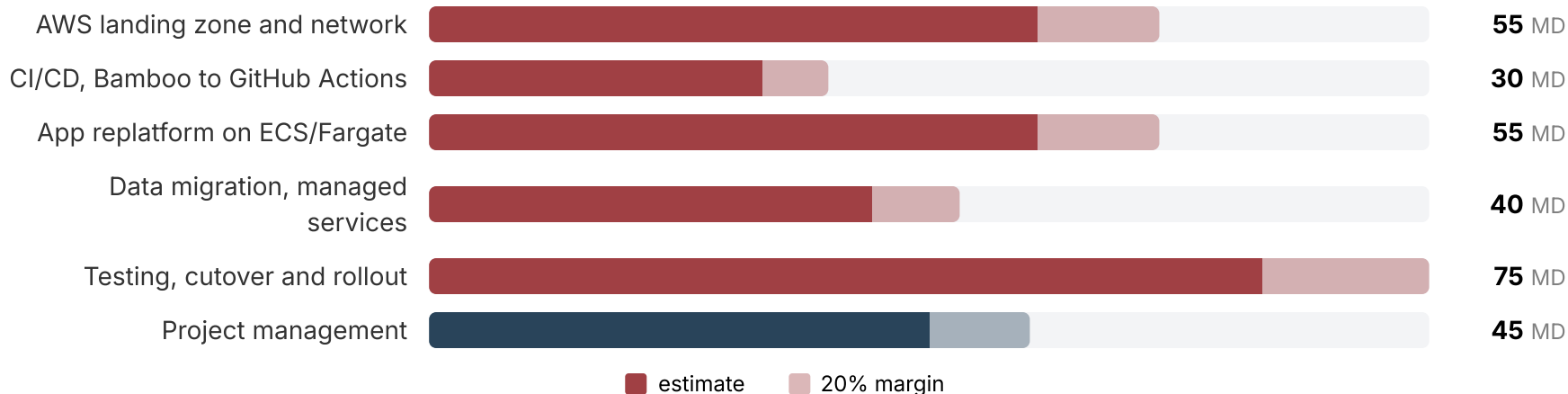
Service continuity

adapting operations and on-call during ramp-up

Each risk has an owner and a mitigation tracked in the migration plan.

Migration effort

One-time effort to replatform onto AWS, CI/CD from Bamboo to GitHub included. Phase 1 only, not the cloud-native refactor.

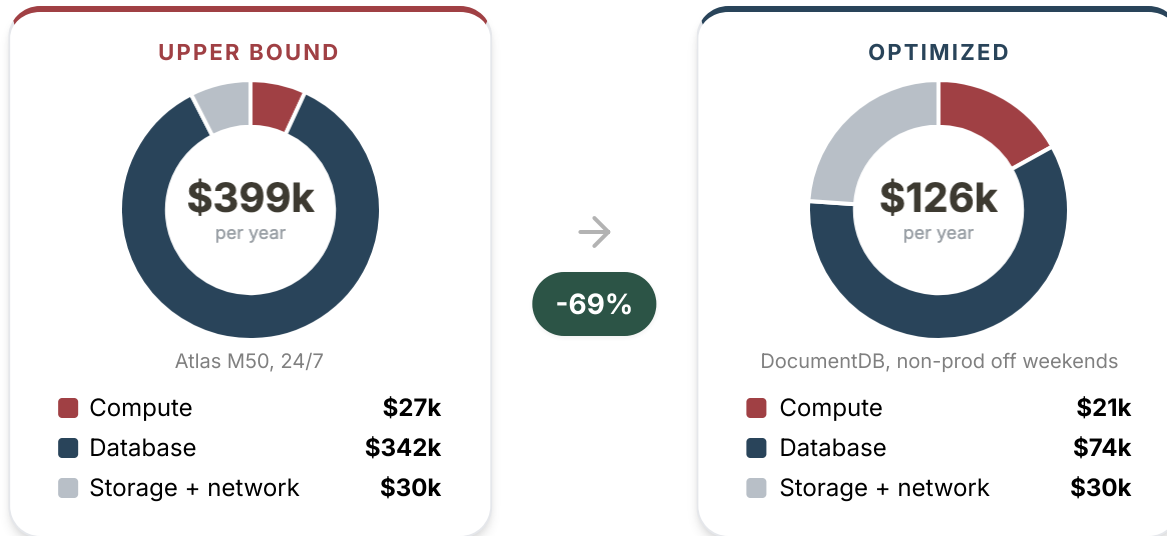


~360
man-days (300 + 20% margin)

~1.6
FTE-year

Cost, milestone and governance

Zurich list prices, 5 environments at 8 vCPU / 32 GB. Atlas is billed per node; DocumentDB non-prod pauses on weekends.



Figures are firm through 2027. For the cloud-native years (2028+), budget a ~25% contingency, so plan ~\$157k for the optimized target.

👤 Run ~1.5 FTE/year

📅 Integration env on AWS, end 2026

👤 Cloud committee

Next steps

What we are asking for today.



Green-light the plan

approve the two-phase migration
and its budget envelope



Chain the ADRs

record the architecture decisions
back to back to validate and start
fast



Integration env by end 2026

running on AWS inside a real
deployment pipeline

Questions and discussion welcome